



Pavlo Hrab

Date of birth: 09.05.1998

Available from 1 December 2026

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website: pavlohrab.com

extended CV: pavlohrab.com/cv

Summary

Computational biologist with a PhD in bioinformatics and microbiology. I build pipelines and open-source tools in Nextflow, Python and R, and use them on large sequencing datasets: metagenomics, transcriptomics, phylogenomics, functional annotation and machine learning. During my PhD I cleaned and analysed 19,720 metagenome-assembled genomes with a pipeline I wrote and released, and ran a year-long cultivation study to isolate the bacteria behind them.

Professional skills

Laboratory	Cultivation and isolation of bacteria; PCR; DNA cloning.
Bioinformatics	Large-scale metagenomics and metatranscriptomics, BGC mining (antiSMASH, BiG-SCAPE, MIBiG), phylogenomics, pangenomics, functional annotation (Pfam, GO, HMMER/pyhmmer)
Programming languages	Python, R, bash, SQL
Statistics & ML	Statistical analysis in R and Python, random forests, clustering, enrichment testing, data cleaning and exploration
Workflows & reproducibility	Nextflow, Conda, Git, working on remote Linux servers, R/Python packaging, Shiny apps

Work experience

Postdoctoral researcher

Wageningen University and Research; Bioinformatics group (Marnix Medema group)

Jan 2026 - Nov 2026

Wageningen, Netherlands

- Mined marine sponge metagenomes for biosynthetic gene clusters, and selected candidates for downstream heterologous expression

PhD Candidate

Wageningen University and Research; Bioinformatics group & Microbiology group
Supervisors: Detmer Sipkema; Marnix Medema

Jan 2022 - Dec 2025,

Wageningen, Netherlands

- Built and released a Nextflow pipeline for MAG quality control and used it to reduce 19,720 public Acidobacteriota genomes to a clean species-level set
- Analysed 488 marine genomes across 16 families to separate host adaptation from shared ancestry, combining phylogenomics, Pfam-based feature matrices and random forest models
- Planned and led a year-long, uninterrupted isolation study of sponge-associated bacteria under unconventional cultivation conditions, uncovering new physiological properties in the isolates; used functional predictions from the metagenomes to design the media
- Supervised 2 MSc and 1 BSc student across computational and molecular work; developed and taught the computational part of an MSc microbiology course
- Presented at 11 multidisciplinary conferences across microbiology, bioinformatics and natural products

Technical Support Specialist

Information System Development

- Reproduced and diagnosed reported bugs against testing servers

May 2021 - Oct 2021,

Lviv, Ukraine

in a regulated, HIPAA-compliant environment

Education

MSc, Ivan Franko National University of Lviv
Major - Genetics and Biotechnology.
(diploma cum laude)

Sep 2019 - Dec 2020
Lviv, Ukraine

BSc, Ivan Franko National University of Lviv
Major - Biology.

Sep 2015 - Jun 2019
Lviv, Ukraine

Selected software

ashelper

github.com/pavlohrab/ashelper

Turns antiSMASH and BiG-SCAPE output into iTOL annotation files python package.

MAG Cleaning Pipeline

github.com/pavlohrab/Mag_cleaning_pipeline

Nextflow pipeline that dereplicate, quality-filter and taxonomically classify set of MAGs CheckM2, GUNC, dRep, GTDB-Tk. Run on 19,720 genomes.

BGCviz

github.com/ostash-group/BGCviz

R/Shiny app comparing biosynthetic gene cluster predictions from six tools on one circular genome plot. R package with tests, vignettes and a hosted demo.

Full list of produced software: pavlohrab.com/tools

Publications

Jackson, S. A., **Hrab, P.**, et al (2026). New insights into the microbiome of the deep-sea sponge Inflatella pellicula and the secondary metabolic potential of metagenome-assembled genomes and the wider microbiome. Microbial genomics, 12(1), 001602. <https://doi.org/10.1099/mgen.0.001602>

Full list: pavlohrab.com/publications

Awards and Honors

Summer School for Natural Products research
August 28, 2025, Ukraine

Invited speaker

European Conference on Marine Natural Products 2025

Oral talk

Pavlo Hrab, et al. "Marine Acidobacteriota as halogenation reservoirs: Discovery of novel biosynthetic gene clusters",
September 29 - October 3, 2025, Piran, Slovenia

Full list of conference presentations: pavlohrab.com/outreach

Languages

English (advanced, C1), **Ukrainian** (native), **Dutch** (beginner, A1)

Hobbies

Reading, water polo, and multiday trekking.